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The responses of bonobos (*Pan paniscus*) to their mirror images: Evidence of self-recognition

This research examined the responses of bonobos (*Pan paniscus*) to their mirror images. Nine bonobos were presented alternately with the reflective and non-reflective sides of a mirror. The apes exhibited considerable interest in the mirror, and immature animals exhibited higher frequencies of contingent action and inactive looking than did adults. Four animals used the mirror to inspect parts of their bodies that were otherwise not visible to them, indicating that bonobos are capable of self-recognition.

It is well-established that orangutans and chimpanzees exhibit behavior indicative of self-recognition (Gallup and Suarez, 1986; Povinelli, Rulf, Landau, & Bierschwale, 1993; Swartz and Evans, 1991). These great apes use mirrors to visually inspect body parts and facial marks that are otherwise not visible to them. There is some controversy as to whether gorillas are capable of self-recognition (Suarez and Gallup, 1981; Patterson, 1984; Parker, Mitchell, and Boccia, 1994; Patterson and Cohn, 1994), and there is no evidence of self-recognition among lesser apes or monkeys (Anderson, 1984).

Bonobos (*Pan paniscus*) are closely related to chimpanzees (*Pan troglodytes*) but differ somewhat in morphology and behavior (Kano, 1982). Bonobos use visual graphic communication systems (Savage-Rumbaugh, McDonald, Sevcik, Hopkins and Rupert, 1986) and exhibit remarkable speech comprehension skills (Savage-Rumbaugh, 1988). However, to the best of our knowledge there have been no quantitative analyses on the responses of bonobos to mirror-image stimulation (but see Hyatt and Hopkins, 1994).

The purpose of this research was to examine the responses of bonobos to their mirror images. Informal observations indicate that a language trained bonobo, Kanzi, exhibits behavior indicative of self-recognition, including the use of a mirror to aid in grooming otherwise non-visible facial regions (Hyatt and Hopkins, personal observations). The present study was undertaken in an effort to provide quantitative data on the use of mirrors in this unique and relatively unstudied species of great ape.

Methods

Subjects and apparatus. The subjects were nine bonobos housed in two social groups in indoor/outdoor cages at the Yerkes Regional Primate Research Center. They included four males and five females that ranged in age from 27 months to an estimated age of 463 months (see Table 1). The apparatus was a mirror that measured 30 x 120 cm. As far as we know all subjects were naive with regard to mirror exposure at the onset of this experiment.

TABLE 1 - Subject information.

Subject	Age in months	Sex	Mirror exposures
Murphy	27	M	3
Brian	43	M	3
Jill	85	F	5
Zalia	100	F	2
Mabruki	114	M	2
Bosondjo	259*	M	3
Lorel	279	F	3
Laura	300	F	3
Linda	463*	F	3

* Age estimated from colony records

Procedure. The animals were tested in their home cage groups. At the onset of each session the mirror was positioned horizontally 50 cm outside the cage. Each session consisted of a 30 minute "mirror-on" condition, when the reflecting side of the mirror faced the subjects, and a 30 minute "mirror-off" condition, when the cardboard backside faced the subjects. The presentation

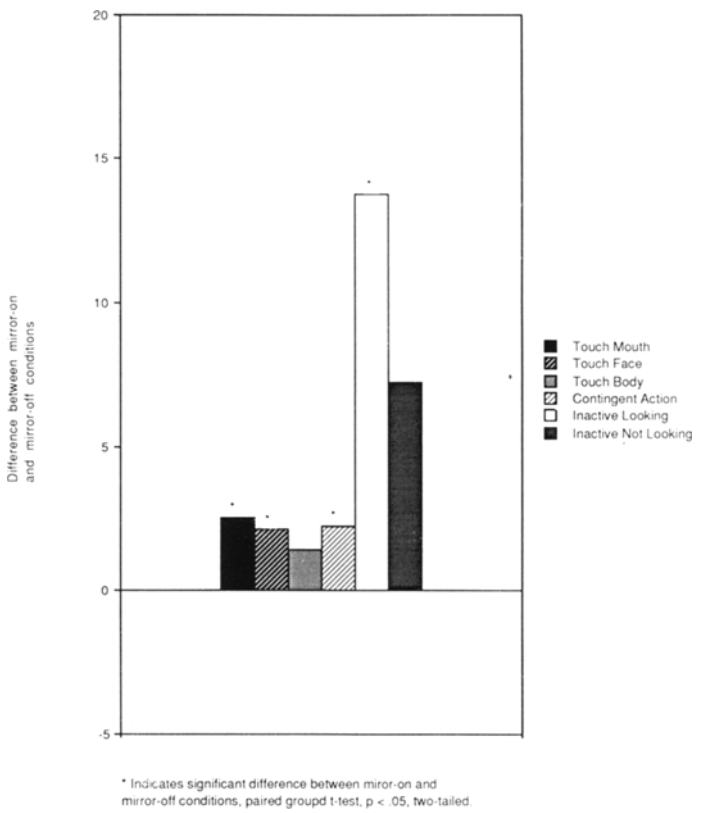


Figure 1 - Differences in the adjusted frequencies of coded behaviors between the mirror-on and mirror-off conditions. A positive value indicates a greater adjusted frequency score in the mirror-on condition.

TABLE 2 - Operational definitions of coded behaviors

Self-directed behavior	
Touch-mouth	Physical contact with the mouth with either the hands or feet.
Touch-face	Physical contact directed toward the face and head region with either the hands or feet. Does not include mouth touching.
Touch body	Physical contact directed toward any region of the body with the exception of the face.
Contingent behavior	
Contingent action	Continued and sustained responses to the mirror which provide response feedback between the subject and mirror. Includes running in front of the mirror, exposure of teeth, shaking, displaying, kicking, clapping, reaching, and lip-smacking.
Inactive visual behavior	
Looking	Visual gaze directed toward the mirror in the absence of self-directed or contingent action.
Not looking	Present but not directing visual gaze toward the mirror.
Other	
Not visible	Subject not visible

order of conditions was counterbalanced across sessions. Between two and five sessions were conducted with each subject. A video camera was positioned behind the mirror and recorded the animals' behavior as they were positioned in front of the apparatus in each condition.

Data were collected from videotape by an observer who noted the frequency and duration of the behavior defined in Table 2. The behavioral taxonomy included self-directed behavior, contingent behavior, inactive visual behavior, and other behavior (see Lewis and Brooks-Gunn, 1979). Thirteen percent of the videotape was randomly selected and scored by two persons to assess observer reliability. This procedure indicated significant inter-rater agreement across the behavioral measures ($r=.92$, $df=8$, $p<.05$). Qualitative notes provided supporting details.

Analysis

The frequency and duration of each coded behavior was determined for each subject. An adjusted frequency score was calculated by dividing the absolute frequency of a behavior by the number of test sessions in which an animal participated. A relative duration score was calculated by dividing the total duration of a behavior by the amount of time that an animal was in view of the camera. The paired groups t-test was used to compare these scores across conditions. Difference scores were calculated by subtracting the adjusted scores obtained in the mirror-on condition from those obtained in the mirror-off condition. A positive value indicated that the adjusted frequency or duration of a behavior was higher in the mirror-on condition than in the mirror-off

condition. The independent groups t-test was used to compare difference scores across age classes. Alpha was set at .05, two-tailed, for all analyses.

Results

Figure 1 provides a summary of the animals' adjusted frequency scores. Overall, adjusted frequency scores were higher for mouth touching, face touching, contingent action, and inactive looking in the mirror-on condition than in the mirror-off condition ($t(8)=2.38$, 2.49, 3.87 and 4.18, respectively for each behavior; mean adjusted frequency score per subject=5.4, 3.2, 3.4, and 19.6 in the mirror-on condition vs. 2.8, 1.0, 1.2, and 5.9 in the mirror-off condition, respectively for each behavior). Adjusted frequency scores did not differ between conditions for body touching and inactive not-looking.

Figure 2 provides a summary of the animals' relative duration scores. Overall, relative duration scores were higher for contingent action and inactive looking in the mirror-on condition than in the mirror-off condition ($t(8)=2.37$ and 4.84, respectively for each behavior, mean relative duration score per subject=5.4 and 44.1 in the mirror-on condition vs. 0.8 and 15.5 in the mirror-

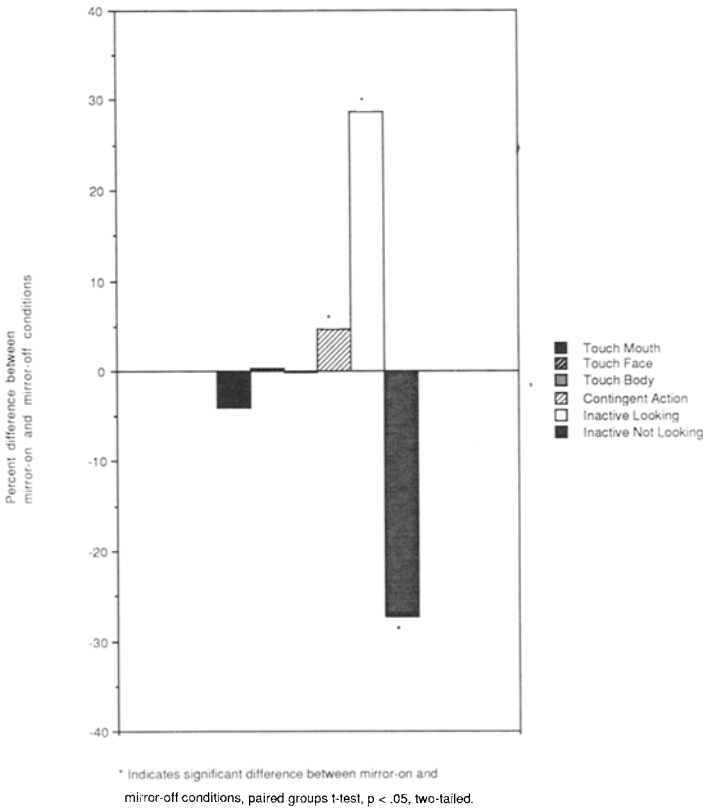


Figure 2 - Differences in the relative durations of coded behaviors between the mirror-on and mirror-off conditions. A positive value indicates a greater relative duration score in the mirror-on condition. A negative value indicates a greater relative duration score in the mirror-off condition.

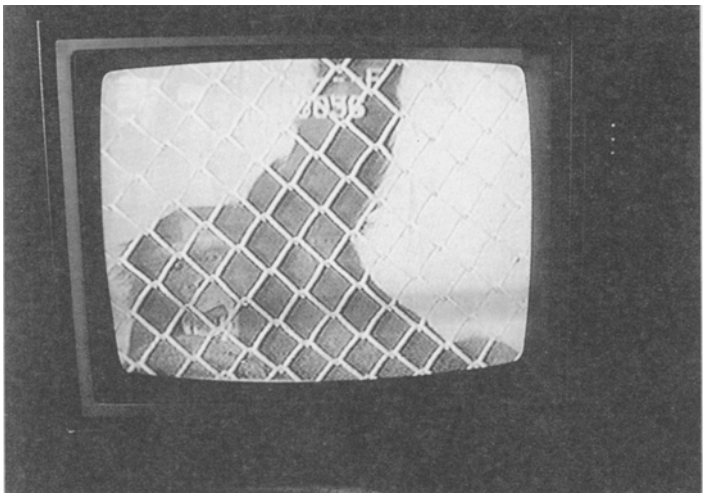


Figure 3 - Bosondjo, and adult male bonobo, using a mirror to aid in the examination of his mouth (Photograph from video by C.W. Hyatt).

off condition, respectively for each behavior). The relative duration score for inactive not-looking was lower in the mirror-on condition than in the mirror-off condition ($t(8)=3.94$; mean relative duration score per subject=43.5 vs. 70.8). Relative duration scores did not differ between conditions for face touching, mouth touching, and body touching.

Immature bonobos (≤114 months of age) exhibited higher adjusted frequency difference scores for contingent action and inactive looking than did adults (≥259 months of age; $t(7)=4.46$ and 2.74, respectively for each behavior). The mean adjusted frequency difference score per subject for contingent action was 3.5 for immature bonobos and 0.7 for adults. The mean adjusted frequency difference score per subject for looking behavior was 19.7 for immature bonobos and 6.3 for adults.

Four animals (Bosondjo, Jill, Laura, and Zalia) exhibited mirror-aided self-inspection. Three of these apes (Bosondjo, Jill and Laura) used the mirror to monitor their behavior as they groomed the inside portions of their mouths (see Figure 3). The fourth animal (Zalia) used the mirror to monitor her behavior as she groomed her eye. Jill exhibited an unusual instance of mirror-aided self-inspection when she contorted her body upside down and used the mirror to view her sexual swelling. It must be noted that these animals did not exhibit these behaviors during the mirror-off condition.

Discussion

Bonobos exhibited considerable interest in their mirror images. Four animals used the mirror to monitor their actions as they inspected body parts that were otherwise not visible to them. These results provide the first quantitative data on the responses of bonobos to their mirror images, and suggest that these apes are capable of self-recognition. Further study will be required to determine if bonobos use mirrors to locate and examine dye marks placed unobtrusively on

their faces (Gallup, 1970), and to guide reaching actions toward hidden objects, as do chimpanzees and monkeys (Anderson, 1986; Menzel, Savage-Rumbaugh, and Lawson, 1985).

Immature bonobos were more interested in their mirror images than were adults. Although small sample size limited our ability to examine the influence of age on self-recognition, we do consider this to be an important area for future study. The youngest subject that used the mirror for self-inspection in this study was Jill, aged 85 months. Further study will be necessary to determine if bonobos exhibit developmental patterns of mirror inspection and self-recognition which are comparable to those of chimpanzees (Lin, Bard, Anderson, 1992; Povinelli, Rulf, Landau, Bierschwale, 1993).

In conclusion, bonobos appear to be capable of self-recognition. Given the close phylogenetic relationship between *Pan paniscus* and *Pan troglodytes*, this is perhaps not surprising. These data provide a starting point for future study on the emergence of self-recognition in bonobos. We believe that future studies with these apes will shed light on the relationship between the emergence of self-recognition and other cognitive processes.

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